

Scattering part 2

From the Born approximation we have the scattering amplitude

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar}\right) V(\mathbf{x}) d\mathbf{x}$$

where $\mathbf{p}$ is momentum transfer

$$\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f$$

and

$$p^2 = |\mathbf{p}|^2 = 4mE(1 - \cos\theta)$$

Let V be a spherically symmetric potential $V(r)$. Show that

$$f(\theta) = -\frac{2m}{\hbar p} \int_0^\infty \sin\left(\frac{pr}{\hbar}\right) V(r) r dr$$

For the Born approximation in polar coordinates we have

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp\left(\frac{ipr \cos\theta}{\hbar}\right) V(r) r^2 \sin\theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$f(\theta) = -\frac{m}{\hbar^2} \int_0^\infty \int_0^\pi \exp\left(\frac{ipr \cos\theta}{\hbar}\right) V(r) r^2 \sin\theta dr d\theta$$

Transform the integral over θ into an integral over u where $u = \cos\theta$ and $du = -\sin\theta d\theta$. The minus sign in du is canceled by interchanging integration limits $\cos 0 = 1$ and $\cos \pi = -1$.

$$f(\theta) = -\frac{m}{\hbar^2} \int_0^\infty \int_{-1}^1 \exp\left(\frac{ipru}{\hbar}\right) V(r) r^2 dr du$$

Solve the integral over u .

$$f(\theta) = -\frac{m}{\hbar^2} \int_0^\infty \left[\frac{\hbar}{ipr} \exp\left(\frac{ipru}{\hbar}\right) \right]_{u=-1}^{u=1} V(r) r^2 dr$$

Cancel r and evaluate limits.

$$f(\theta) = -\frac{m}{\hbar p} \int_0^\infty \left[-i \exp\left(\frac{ipr}{\hbar}\right) + i \exp\left(-\frac{ipr}{\hbar}\right) \right] V(r) r dr$$

Recalling that

$$2 \sin x = -ie^{ix} + ie^{-ix}$$

we have

$$f(\theta) = -\frac{2m}{\hbar p} \int_0^\infty \sin\left(\frac{pr}{\hbar}\right) V(r) r dr$$

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